



Auxiliary devices



Amplifier What about the amplifier?

- ▶ Supported using *auxiliary devices*
- ▶ Register a `struct snd_soc_aux_dev` array using the `.aux_dev` and `.num_aux_devs` fields of the registered `struct snd_soc_card`
- ▶ This will expose the auxiliary devices control widgets as part of the sound card
- ▶ There is a driver for simple amplifiers driven by a single GPIO, `simple-amplifier`
 - [Documentation/devicetree/bindings/sound/simple-audio-amplifier.yaml](#)
 - [sound/soc/codecs/simple-amplifier.c](#)



Auxiliary devices

sound/soc/samsung/neo1973_wm8753.c

```
static struct snd_soc_aux_dev neo1973_aux_devs[] = {
    {
        .name = "dfbmcs320",
        .codec_name = "dfbmcs320.0",
    },
};

static struct snd_soc_card neo1973 = {
    .name = "neo1973",
    .owner = THIS_MODULE,
    .dai_link = neo1973_dai,
    .num_links = ARRAY_SIZE(neo1973_dai),
    .aux_dev = neo1973_aux_devs,
    .num_aux_devs = ARRAY_SIZE(neo1973_aux_devs),
};
```



simple-amplifier - example 1

arch/arm64/boot/dts/allwinner/sun50i-a64-pinebook.dts

```
speaker_amp: audio-amplifier {  
    compatible = "simple-audio-amplifier";  
    VCC-supply = <&reg_ldo_io0>;  
    enable-gpios = <&pio 7 7 GPIO_ACTIVE_HIGH>; /* PH7 */  
    sound-name-prefix = "Speaker Amp";  
};
```

```
&sound {  
    status = "okay";  
    simple-audio-card,aux-devs = <&codec_analog>, <&speaker_amp>;  
    simple-audio-card,widgets = "Microphone", "Internal Microphone Left",  
                                "Microphone", "Internal Microphone Right",  
                                "Headphone", "Headphone Jack",  
                                "Speaker", "Internal Speaker";  
    simple-audio-card,routing =  
        "Left DAC", "AIF1 Slot 0 Left",  
        "Right DAC", "AIF1 Slot 0 Right",  
        "Speaker Amp INL", "LINEOUT",  
        "Speaker Amp INR", "LINEOUT",  
        "Internal Speaker", "Speaker Amp OUTL",  
        "Internal Speaker", "Speaker Amp OUTR",  
        "Headphone Jack", "HP",
```



simple-amplifier - example 2

```
dio2133: analog-amplifier {
    compatible = "simple-audio-amplifier";
    sound-name-prefix = "AU2";
    VCC-supply = <&hdmi_5v>;
    enable-gpios = <&gpio GPIOH_5 GPIO_ACTIVE_HIGH>;
};

sound {
    compatible = "amlogic,gx-sound-card";
    model = "GXL-LIBRETECH-S905X-CC";
    audio-aux-devs = <&dio2133>;
    audio-widgets = "Line", "Lineout";
    audio-routing = "AU2 INL", "ACODEC LOLN",
                   "AU2 INR", "ACODEC LORN",
                   "Lineout", "AU2 OUTL",
                   "Lineout", "AU2 OUTR";
}
```

Audio is routed through **AU2**, the amplifier.



Input Muxing

- ▶ There may be a muxer on the analog input lines.
- ▶ If controlled using a gpio, the [simple-mux](#) driver is available.
- ▶ It exposes two inputs: “IN1” and “IN2” and one output, “OUT”.
- ▶ The device tree binding allows to provide a prefix to make the routes specific.
 - [Documentation/devicetree/bindings/sound/simple-audio-mux.yaml](#)
 - [sound/soc/codecs/simple-mux.c](#)



simple-mux example

```
mic_mux: mic-mux {  
    compatible = "simple-audio-mux";  
    pinctrl-names = "default";  
    pinctrl-0 = <&pinctrl_micse1>;  
    mux-gpios = <&gpio5 5 GPIO_ACTIVE_LOW>;  
    sound-name-prefix = "Mic Mux";  
};
```

- ▶ This exposes routes between Mic Mux IN1 and Mic Mux IN2 to Mic Mux OUT.
- ▶ This route is controlled by gpio5 5.
- ▶ A control named Mic Mux Muxer will be exposed to userspace.



simple-mux example

arch/arm64/boot/dts/freescale/imx8mq-librem5-devkit.dts

```
sound {
    compatible = "simple-audio-card";
    pinctrl-names = "default";
    pinctrl-0 = <&pinctrl_hpdet>;
    simple-audio-card,aux-devs = <&speaker_amp>, <&mic_mux>;
    simple-audio-card,name = "Librem 5 Devkit";
    simple-audio-card,format = "i2s";
    simple-audio-card,widgets =
        "Microphone", "Builtin Microphone",
        "Microphone", "Headset Microphone",
        "Headphone", "Headphones",
        "Speaker", "Builtin Speaker";
    simple-audio-card,routing =
        "MIC_IN", "Mic Mux OUT",
        "Mic Mux IN1", "Headset Microphone",
        "Mic Mux IN2", "Builtin Microphone",
        "Mic Mux OUT", "Mic Bias",
        "Headphones", "HP_OUT",
        "Builtin Speaker", "Speaker Amp OUTR",
        "Speaker Amp INR", "LINE_OUT";
    simple-audio-card,hp-det-gpio = <&gpio3 20 GPIO_ACTIVE_HIGH>;

    simple-audio-card,cpu {
        sound-dai = <&sai2>;
    }
}
```



simple-mux example

```
};

simple-audio-card,codec {
    sound-dai = <&sgtl5000>;
    clocks = <&clk IMX8MQ_CLK_SAI2_ROOT>;
    frame-master;
    bitclock-master;
};

};
```